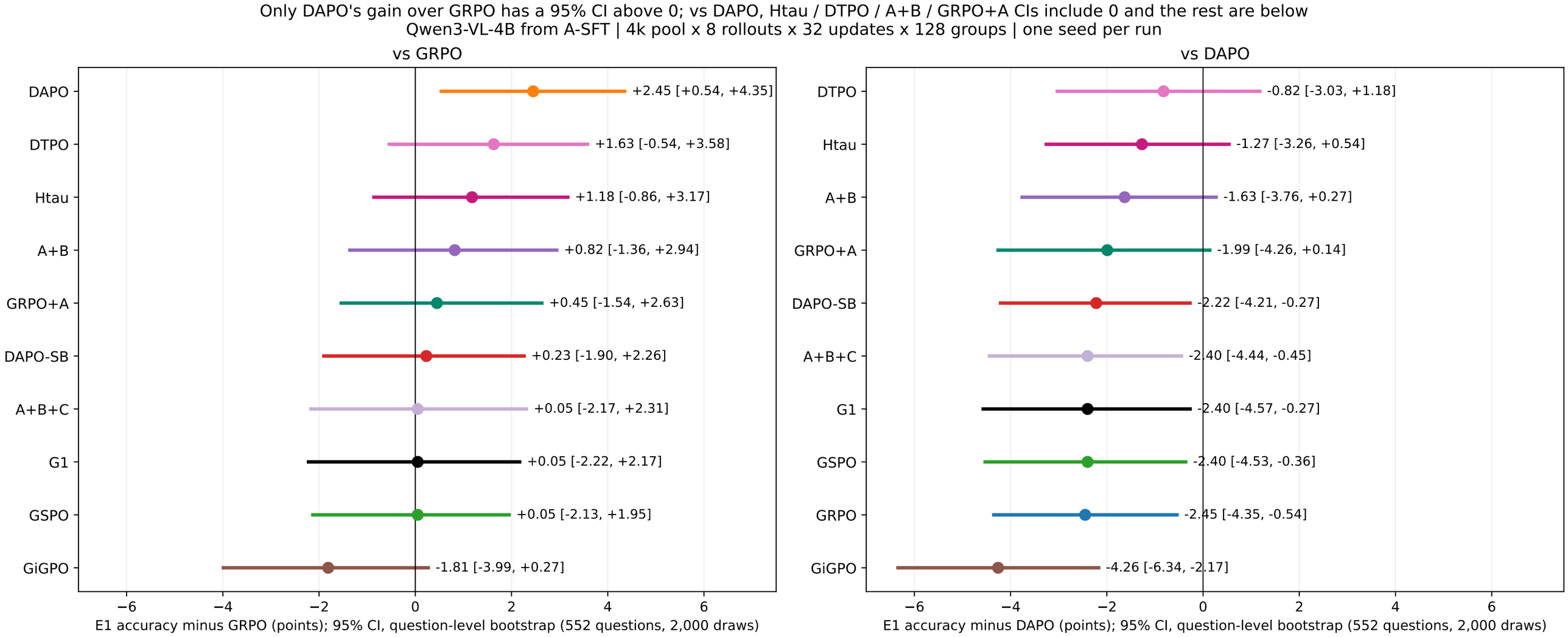
Only DAPO's gain over GRPO has a 95% CI above 0; vs DAPO, Htau / DTPO / A+B / GRPO+A CIs include 0 and the rest are below
Qwen3-VL-4B from A-SFT | 4k pool x 8 rollouts x 32 updates x 128 groups | one seed per run
vs GRPO
DAPO
DTPO
Htau
A+B
GRPO+A
DAPO-SB
A+B+C
G1
GSPO
GiGPO
+2.45 [+0.54, +4.35]
+1.63 [-0.54, +3.58]
+1.18 [-0.86, +3.17]
+0.82 [-1.36, +2.94]
+0.45 [-1.54, +2.63]
+0.23 [-1.90, +2.26]
+0.05 [-2.17, +2.31]
+0.05 [-2.22, +2.17]
+0.05 [-2.13, +1.95]
-1.81 [-3.99, +0.27]
−6
−4
−2
0
2
4
6
E1 accuracy minus GRPO (points); 95% CI, question-level bootstrap (552 questions, 2,000 draws)
vs DAPO
DTPO
Htau
A+B
GRPO+A
DAPO-SB
A+B+C
G1
GSPO
GRPO
GiGPO
-0.82 [-3.03, +1.18]
-1.27 [-3.26, +0.54]
-1.63 [-3.76, +0.27]
-1.99 [-4.26, +0.14]
-2.22 [-4.21, -0.27]
-2.40 [-4.44, -0.45]
-2.40 [-4.57, -0.27]
-2.40 [-4.53, -0.36]
-2.45 [-4.35, -0.54]
-4.26 [-6.34, -2.17]
−6
−4
−2
0
2
4
6
E1 accuracy minus DAPO (points); 95% CI, question-level bootstrap (552 questions, 2,000 draws)